

Multi-Layer Dos Detection and Severity Scoring System Using
Hybrid ML and DL Approaches.

Project ID- R26-CS-016

Project Proposal Report
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B.Sc. (Hons) Degree in Information Technology Specialized in
cybersecurity

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March 2026

DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Denial of Service (DoS) attacks, particularly Internet Control Message Protocol (ICMP) flood attacks, pose a critical threat to network availability by overwhelming target systems with malicious Echo Request packets, exhausting bandwidth and processing resources. While existing detection systems demonstrate reasonable classification accuracy, they predominantly produce binary outcomes attack or normal without quantifying the relative severity or real-world impact of detected attacks. This limitation hinders Security Operations Centre (SOC) analysts from making fast, risk-informed mitigation decisions.

This research proposes and implements a severity-aware ICMP flood detection component as part of a broader Multi-Layer DoS Detection and Severity Scoring System (Project R26-CS-016). The proposed system extracts ICMP-specific Layer 3 network traffic features including packet rate, Echo Request count, source IP diversity, and time-window statistics from publicly available benchmark datasets (CICDDoS2019 and CAIDA). A hybrid detection approach combining Machine Learning models Random Forest and Support Vector Machine (SVM) with Deep Learning models Convolutional Neural Network (CNN) or Long Short-Term Memory (LSTM) is employed to accurately classify ICMP flood traffic.

The core contribution of this work is a **CVSS-inspired weighted severity scoring engine** that quantifies the operational impact of detected ICMP flood attacks. Instead of producing only binary detection results, the proposed system computes a severity score by multiplying the detected ICMP attack packet count by an impact weight of **1.5**, where the weight reflects the relative availability impact of ICMP flooding in terms of bandwidth saturation and packet-processing overhead. The computed score is then mapped into four severity levels: **Low, Medium, High, and Critical**. The output is presented as a structured **ICMP Attack Severity Ranking Table**, which is further forwarded to a system-wide **Final Severity Aggregator**. Target performance metrics include detection accuracy of at least **90%**, an **F1-score above 88%**, and a **false positive rate below 5%**. By integrating quantified severity analysis with ICMP flood detection, this work addresses a clear limitation in existing research and supports more effective prioritization of network threat response.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude and deepest appreciation to all those who have contributed, supported, and guided me throughout the development of this research proposal.

First and foremost, I am profoundly grateful to my research supervisor and co-supervisor at SLIIT University for their expert guidance, continuous encouragement, and constructive feedback at every stage of this research. Their in-depth knowledge in the fields of cybersecurity, network security, and machine learning has been invaluable in shaping the research direction, refining the problem statement, and ensuring the academic rigor of this proposal. Their patience and dedication in reviewing drafts and providing timely direction have been truly instrumental in bringing this work to its current form.

I would also like to extend my sincere appreciation to my fellow research team members Farhan R (IT22284334), Kezothran G (IT22284334), and Kajan S (IT22122346) for their outstanding collaboration, mutual support, and shared commitment to achieving a high-quality research outcome. The collective effort, open communication, and complementary expertise of each team member across the SYN flood, UDP flood, and fragmentation attack components have greatly strengthened the overall multi-layer detection framework.

I am also thankful to the Faculty of Computing at SLIIT University for providing access to essential academic resources, research infrastructure, relevant datasets, and a stimulating intellectual environment that enabled the successful development of this research.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
A2M	Adaptive Multi-Layered Defence Mechanism
AWS	Amazon Web Services
CAIDA	Center for Applied Internet Data Analysis
CIC	Canadian Institute for Cybersecurity
CICDDoS2019	Canadian Institute for Cybersecurity DDoS Dataset 2019
CIC-IDS2017	Canadian Institute for Cybersecurity Intrusion Detection System Dataset 2017
CNN	Convolutional Neural Network
CSV	Comma-Separated Values
DDoS	Distributed Denial of Service
DENFIS	Dynamic Evolving Neural Fuzzy Inference System
DL	Deep Learning
DoS	Denial of Service
DSR	Design Science Research
EC2	Elastic Compute Cloud
F1	F1 Score (Harmonic Mean of Precision and Recall)
IAS	Information Assurance and Security
ICMP	Internet Control Message Protocol
IDS	Intrusion Detection System
IEEE	Institute of Electrical and Electronics Engineers
IP	Internet Protocol
IPS	Intrusion Prevention System
IoT	Internet of Things
ISP	Internet Service Provider
ITU	International Telecommunication Union
k-NN	k-Nearest Neighbours
L3	Layer 3 (Network Layer)
LKR	Sri Lankan Rupee
LSTM	Long Short-Term Memory
ML	Machine Learning
MTTR	Mean Time to Respond
NDP	Neighbour Discovery Protocol
PCAP	Packet Capture
PPDIOO	Prepare, Plan, Design, Implement, Operate, Optimize
RF	Random Forest
RMSE	Root Mean Square Error
SaaS	Software as a Service
SDG	Sustainable Development Goal
SDN	Software-Defined Networking
SIEM	Security Information and Event Management
SLA	Service Level Agreement
SLIIT	Sri Lanka Institute of Information Technology
SOC	Security Operations Centre

1. INTRODUCTION

1.1 Background and Context

Denial of Service (DoS) attacks remain one of the most critical and persistent cybersecurity threats in modern network environments. These attacks disrupt service availability by overwhelming target systems with malicious traffic, rendering them inaccessible to legitimate users. Among the various types of DoS attacks, Internet Control Message Protocol (ICMP) flood attacks represent a widely exploited attack vector. An ICMP flood attack involves transmitting an abnormally large volume of ICMP Echo Request (ping) packets to a target system, exhausting its processing capacity and bandwidth, thereby degrading or completely disrupting network services.

Organizations across sectors including enterprises, cloud service providers, Internet Service Providers (ISPs), financial institutions, healthcare systems, and government networks are primary victims due to their dependence on continuous network availability. According to the International Telecommunication Union (ITU), there were 5.38 billion internet users globally as of 2021, highlighting the scale of the digital infrastructure that must be protected [1]. The increasing interconnection of devices, particularly within IoT environments, has further expanded the attack surface available to malicious actors.

ICMP flood attacks exploit the fundamental operation of the ICMP protocol, which is natively used for diagnostic purposes such as network path testing, error reporting, and status queries. The attacker sends massive volumes of ICMP_ECHO_REQUEST packets to the victim; the victim attempts to reply to each request, consuming CPU cycles, memory, and network bandwidth [2]. In the Smurf variant of ICMP flooding, the attacker directs ICMP ECHO requests to broadcast addresses while spoofing the victim's IP address, causing many machines to reply simultaneously and amplifying the attack effect.

Current DoS detection systems largely operate on binary classification outcomes either classifying traffic as an attack or normal and fail to differentiate the relative severity and real-world impact of different attack types. This limitation leads to inefficient prioritization of mitigation actions by security operations center (SOC) analysts and network administrators. A network security system that not only detects ICMP flood attacks but also quantifies their severity using a weighted scoring mechanism would provide significant value in real-world security operations.

This research component aligns with the IAS Information Assurance & Security research cluster at SLIIT, focusing on securing digital infrastructures through intelligent and adaptive security mechanisms. It supports SDG 9 (Industry, Innovation, and Infrastructure) by strengthening resilient digital infrastructure and SDG 16 (Peace, Justice, and Strong Institutions) by enhancing cyber resilience.

1.2 Literature Review

Extensive research has been conducted on DoS and DDoS attack detection, mitigation, and analysis. This section reviews key studies related to ICMP flood attacks, machine learning and deep learning-based intrusion detection, and severity scoring approaches relevant to network threat prioritization.

1.2.1 ICMP Flood Attacks: Mechanisms and Impact

Harshita [3] provides foundational analysis of ICMP flood DDoS attacks, outlining how attackers exploit ICMP protocol properties including the packet format, TTL parameters, and echo request/reply to mechanics. The study surveys 50 websites across banking, education, government, gaming, and private sectors, demonstrating varying levels of vulnerability to ICMP packet manipulation. Key findings indicate that several critical infrastructure sites remain vulnerable to increased packet sizes, particularly in the range of 35,000 bytes, which can trigger flood conditions.

1.2.2 SDN-Based ICMP Detection and Mitigation

Kartadie et al. [1] propose an SDN and sFlow-RT based ICMP flood detection and mitigation system following the PPDIIO lifecycle methodology. Their implementation uses OpenFlow switches and sFlow agents to monitor ICMP traffic patterns in real time. Results demonstrate a significant reduction in ICMP packet volume from 311,130 packets to 99 packets over 120 seconds (99.97% decrease), with latency improvements of approximately 80%. However, the system relies on predefined threshold-based detection (100 bytes/second) rather than ML-based intelligence, limiting its adaptability to novel attack patterns.

1.2.3 Fuzzy and Neural Approaches for ICMPv6 Detection

Saad et al. [4] propose ICMPv6 flood attack detection using the Dynamic Evolving Neural Fuzzy Inference System (DENFIS) algorithm in IPv6 networks. A dataset of 2000 records (attack rates 1000–1500 pings, normal rates 10–15 pings) is used with an 80/20 train/test split. The classifier achieves a Root Mean Square Error (RMSE) of 0.26, demonstrating effective detection of ICMPv6 floods. The study highlights that ICMPv6 cannot be disabled in IPv6 networks (unlike ICMPv4), making robust detection especially critical for next-generation networks.

1.2.4 Ensemble Learning and SVM for ICMP Detection in IoT

Radhakrishnan et al. [5] present a hybrid ensemble learning framework using Decision Tree, Random Forest, k-Nearest Neighbours (k-NN), and Support Vector Machine (SVM) as a meta-classifier for ICMP flood detection in IoT networks. Evaluated on the CIC-IDS2017 dataset, the full model achieves 95% accuracy, 94% precision, 96% recall, 95% F1 score, a 3% false positive rate, and only 20 MS computational overhead. An ablation study confirms that combining all three base classifiers outperforms any individual or pairwise combination. The study validates that ensemble approaches are superior to single-classifier solutions for ICMP detection tasks.

1.2.5 Machine Learning for DoS Detection

Sabu et al. [6] demonstrate the use of ML-based intrusion detection systems for DoS attacks including SYN flood, UDP flood, and ICMP flood. Results using Random Forest and SVM on benchmark datasets show high detection accuracy across multiple attack categories. The research demonstrates that structured statistical features extracted from network traffic are well-suited to ML classifiers for attack classification tasks.

1.2.6 Deep Learning for Real-Time DDoS Detection

Hernandez et al. [2] investigate real-time DDoS detection in high-speed networks using deep learning models including LSTM networks and CNN-LSTM hybrid architectures. Their results show that DL models effectively capture temporal traffic patterns and sequential attack behaviors that are difficult for traditional ML to model. The CNN-LSTM hybrid demonstrates superior performance in high-speed environments, making it highly relevant for the deep learning component of the proposed system.

1.2.7 Multi-Layered Defence Mechanisms

Sutradhar and Sarkar [7] propose the A2M adaptive multi-layered defence mechanism against DDoS attacks, demonstrating that layered defence architectures operating across network, transport, and application layers provide improved resilience against multi-vector attacks. This supports the multi-layer design rationale of the proposed system, which targets Layer 3 (ICMP) and Layer 4 (TCP, UDP) attack types.

1.2.8 AdaBoost for SYN and UDP Detection

Syafiuddin et al. [8] apply AdaBoost ensemble techniques for detecting SYN flood and UDP lag attacks, achieving high detection accuracy. The study reinforces the value of ensemble methods for network attack detection and provides a comparative baseline for evaluating ML models across different DoS attack types.

1.2.9 CVSS-Based Vulnerability Severity Assessment

Wang et al. [9] present a vulnerability assessment framework that integrates the **Common Vulnerability Scoring System (CVSS)** with a Bayesian network model to evaluate cybersecurity risks. The study highlights how CVSS metrics based on **confidentiality, integrity, and availability (CIA)** can quantify the severity of vulnerabilities and support prioritization of security responses. The results demonstrate that combining CVSS scoring with probabilistic models improves the accuracy of vulnerability evaluation, particularly in environments where security data is incomplete.

1.2.10 CVSS Framework for Standardized Vulnerability Scoring

Mell et al. [10] introduce the **Common Vulnerability Scoring System (CVSS)** as a standardized framework designed to measure the severity of software vulnerabilities. The framework categorizes metrics into **base, temporal, and environmental groups**, allowing security analysts to generate consistent vulnerability severity scores across different systems. This standardized scoring mechanism enables organizations to prioritize security mitigation strategies and improve vulnerability management practices.

1.2.11 Evaluation of CVSS Generations and Limitations

Horváth et al. [11] analyze different generations of the CVSS framework and discuss their effectiveness in vulnerability severity evaluation. The study explains how CVSS scoring provides a structured method for assigning numerical severity ratings to vulnerabilities, helping security professionals prioritize responses according to potential impact. However, the research also highlights several limitations in the scoring system, such as ambiguity in certain impact categories and difficulties in accurately representing real-world attack consequences.

1.3 Research Gap

Despite significant advances in DoS and ICMP flood detection, the existing literature reveals a persistent and critical gap: no current system integrates attack detection with a quantified, impact-based severity scoring mechanism. Specifically:

- Existing detection systems produce binary outputs (attack / normal) without indicating attack urgency or real-world impact level [1] [5].
- No published work assigns differentiated impact weights to ICMP flood subtypes to rank their relative danger to network infrastructure.
- Systems such as those by Kartadie et al. [1] rely on static threshold-based detection that cannot adapt to evolving attack patterns without ML intelligence.
- While ensemble ML/DL approaches achieve high detection accuracy [5] [2], they do not provide risk-prioritized output to guide SOC mitigation workflows.
- The DENFIS-based approach [4] targets ICMPv6 in academic testbeds, lacking practical applicability to general enterprise network environments.

This research addresses these gaps by introducing a severity-aware ICMP flood detection component that extends beyond binary attack classification to quantify the operational impact of detected attacks. The proposed system employs a **CVSS-inspired weighted severity scoring engine**, where ICMP flood traffic is detected into a severity score using packet intensity and an **ICMP-specific impact weight**. The resulting score is mapped into prioritized severity levels to support faster and more informed mitigation decisions. In this way, the proposed approach improves the practical usefulness of ICMP flood detection for SOC analysts by linking detection results with attack urgency and response priority.

2.OBJECTIVE

2.1 Main Objective

To design and implement a severity-aware ICMP flood detection component that accurately detects, classifies, and quantifies ICMP flood attacks using hybrid Machine Learning and Deep Learning techniques, and generates a CVSS-inspired severity score to support prioritized mitigation and decision-making within the proposed multi-layer DoS detection framework.

2.2 Specific Objectives

1. To extract and analyze relevant ICMP traffic features including packet rate, ICMP echo request frequency, source IP diversity, and time-window statistics from network traffic datasets (CICDDoS2019 and CAIDA).
2. To implement and evaluate hybrid ML models (Random Forest and SVM) for structured ICMP traffic feature classification, achieving a target detection accuracy of at least 90%.
3. To implement and evaluate deep learning models (CNN or LSTM) for capturing temporal ICMP traffic patterns and improving detection performance in high-volume traffic conditions.
4. To develop and validate a **CVSS-inspired severity scoring engine** that multiplies the detected ICMP attack packet count by an ICMP-specific impact weight of 1.5 to generate a quantified severity score and maps the resulting score into the severity levels Low, Medium, High, and Critical.
5. To generate a structured ICMP attack severity ranking table that enables SOC analysts to assess and prioritize ICMP flood attack responses based on computed severity levels.
6. To validate the severity scoring mechanism against attack impact indicators including traffic volume, persistence duration, and system resource utilization metrics.

3.METHODOLOGY

3.1 Research Approach

This research adopts a Design Science Research (DSR) methodology, focused on constructing and evaluating an artefact a severity-aware ICMP flood detection system to solve a practically relevant cybersecurity problem. The approach follows iterative design, implementation, and evaluation cycles, using experimental evaluation on publicly available benchmark datasets.

3.2 Technical Methods and Algorithms

Phase 1: Data Collection, Preprocessing, and Feature Extraction

Network traffic from the CICDDoS2019 and CAIDA datasets will be preprocessed to extract Layer 3 ICMP-relevant features using time-window analysis. Key features include:

- ICMP packet rate
- ICMP Echo Request count and ICMP Echo Reply count
- Source IP address diversity (unique source IPs per window)
- Packet size distribution and inter-arrival time statistics
- Protocol flags and ICMP type/code fields

Python libraries (Pandas, NumPy, Scikit-learn) and network analysis utilities (Wireshark, Scapy) will be used for feature extraction and preprocessing.

Dataset Preparation and Handling

- Before training the detection models, the collected datasets will undergo a structured preprocessing and preparation process. First, the combined dataset will be divided into **training and testing subsets using a 70:30 split**, where 70% of the data will be used for model training and 30% will be used for performance evaluation. To ensure fair evaluation, the split will preserve the distribution of normal and attack traffic samples.
- Since network intrusion datasets often contain **class imbalance between normal traffic and attack instances**, appropriate imbalance handling techniques will be applied. This may include **random under-sampling of majority classes or synthetic over-sampling techniques such as SMOTE**, ensuring balanced representation of ICMP flood attacks during model training.
- In addition, several preprocessing steps will be performed to improve model performance. These include **removal of incomplete or corrupted records, normalization of numerical traffic features, encoding of categorical attributes such as ICMP type/code fields, and time-window aggregation of packet statistics**. These preprocessing steps ensure that the dataset is clean, consistent, and suitable for machine learning and deep learning model training.

Phase 2: Hybrid ML/DL Detection Models

To detect ICMP flood attacks effectively, the proposed system utilizes both Machine Learning (ML) and Deep Learning (DL) techniques. These models are designed to analyze extracted network traffic features and classify traffic as either normal or malicious.

Detection Models

- Machine Learning models including **Random Forest (RF)** and **Support Vector Machine (SVM)** will be implemented as baseline classifiers using structured ICMP traffic features extracted during the preprocessing phase. These models are widely used in network intrusion detection due to their interpretability, computational efficiency, and strong performance in classification tasks.
- In addition, **Deep Learning models such as Convolutional Neural Networks (CNN) or Long Short-Term Memory (LSTM)** networks will be implemented to capture temporal patterns and sequential characteristics in network traffic. Deep learning approaches are particularly suitable for detecting complex traffic patterns and high-volume attack behavior that may not be easily captured by traditional ML models.

Model Comparison and Hybrid Evaluation Strategy

- To evaluate the effectiveness of the proposed hybrid detection approach, a structured comparison strategy will be applied. Each model (RF, SVM, CNN, and LSTM) will first be trained and evaluated independently using the same training and testing datasets to ensure fair comparison. The performance of the models will be assessed using standard intrusion detection metrics including **detection accuracy, precision, recall, F1-score, and false positive rate**.
- Following this evaluation, the results of the ML and DL models will be compared with the performance of the hybrid ML–DL detection approach. This comparison will determine whether combining ML-based structured feature classification with DL-based temporal pattern recognition improves ICMP flood detection performance and enhances the robustness of the overall detection framework.

All models will be evaluated under the same experimental conditions to ensure fair and consistent performance comparison.

Phase 3: ICMP Packet Counting

Detected ICMP attack packets will be counted within defined time windows. Counts will be disaggregated by attack subtype where applicable (standard ICMP flood, Smurf attack variants). Counting is performed independently by the detection window to enable time-series severity tracking.

Phase 4: Severity Scoring Engine

- The core contribution of this component is a **CVSS-inspired severity scoring mechanism** that quantifies the operational impact of detected ICMP flood attacks. Instead of producing only binary detection results, the proposed system converts attack intensity into a numerical severity score that reflects the potential disruption to network availability.

The severity score is calculated using the following formulation:

- Severity Score = ICMP Packet Count $\times$ Impact Weight
- The ICMP impact weight **1.5** represents a moderate availability impact level within the proposed severity model. ICMP flood attacks primarily cause bandwidth saturation and packet-processing overhead rather than protocol state exhaustion, which typically occurs in attacks such as SYN floods. Therefore, ICMP flooding is considered to have **moderate operational impact relative to other volumetric DoS attacks**. Based on comparative observations reported in prior DoS detection and multi-layer defense studies, a coefficient value of **1.5** is assigned to reflect this moderate-impact category. The weight functions as an **initial severity coefficient** and will be empirically validated during system evaluation by comparing computed severity scores with attack intensity indicators such as packet rate, traffic persistence duration, and system resource utilization.
- The selected weight serves as an initial severity coefficient, and its effectiveness will be validated through empirical evaluation using attack traffic intensity and network performance indicators.
- This scoring mechanism enables the system to **translate raw attack traffic measurements into actionable security intelligence**, supporting prioritized response decisions in Security Operations Center (SOC) environments.

Phase 5: Severity Classification

After computing the severity score, the system maps the score into predefined severity levels representing the urgency and potential operational impact of the attack.

The classification framework consists of four severity levels:

Severity Level	Score Range	Recommended Action
Low	0 – 500	Log and monitor
Medium	501 – 1500	Alert SOC analyst
High	1501 – 3000	Initiate mitigation
Critical	> 3000	Immediate response required

Table 1 Severity Classification

3.3 Tools and Platforms

This classification enables SOC analysts to quickly interpret attack intensity and prioritize mitigation actions, ensuring that high-impact attacks receive immediate attention while lower-impact anomalies are monitored.

Tool / Platform	Purpose	Justification
Python 3.x	ML/DL implementation, data processing	Industry standard for ML; rich library ecosystem
Scikit-learn	Random Forest, SVM implementation	Optimized ML algorithms with good documentation
TensorFlow / Keras	CNN / LSTM deep learning models	Industry standard DL framework
Wireshark / Scapy	Packet capture and feature extraction	Network traffic analysis and ICMP feature extraction
CICDDoS2019 Dataset	Training and testing data	Publicly available, labelled, includes ICMP flood traffic
CAIDA Dataset	Supplementary validation data	Real-world anonymized network traffic for validation
Jupyter Notebook	Experimentation and analysis	Supports iterative ML/DL development and visualization

Table 2 tools

3.4 Data Collection and Ethical Considerations

This project uses publicly available benchmark datasets (CICDDoS2019 and CAIDA) that contain labelled network traffic records covering ICMP flood attack scenarios. No real-world attack execution against live systems will be performed. The research does not involve human participants, personal data, or sensitive identifiable information. All experiments operate on synthetic or anonymized traffic data within controlled simulation environments. This research strictly adheres to ethical guidelines, focusing on defensive cybersecurity mechanisms and responsible data usage.

3.5 Validation Strategy and Performance Metrics

In addition to detection performance, the proposed severity scoring mechanism will be quantitatively evaluated by comparing computed severity scores with attack impact indicators such as ICMP packet rate, attack duration, CPU usage, and bandwidth consumption. Correlation analysis will be used to determine whether higher severity scores correspond to greater operational impact. This helps verify that the scoring mechanism reflects actual attack severity, not only detection output.

Metric	Description	Target
Detection Accuracy	Percentage of correctly classified traffic instances	$\geq 90\%$
Precision	Ratio of true attack detections to total predicted attacks	$\geq 88\%$
Recall (Sensitivity)	Ratio of detected attacks to total actual attacks	$\geq 88\%$
F1 Score	Harmonic mean of Precision and Recall	$\geq 88\%$
False Positive Rate	Percentage of normal traffic misclassified as attacks	$\leq 5\%$
Severity Score Correlation	Correlation between computed severity scores and attack impact indicators such as packet rate, attack duration, CPU usage, and bandwidth usage.	Strong positive correlation

Table 3 Metric

3.6 Limitations and Assumptions

The proposed ICMP flood detection and severity scoring framework is developed under several assumptions and practical limitations. The system assumes that the CICDDoS2019 and CAIDA datasets are sufficiently representative of real-world ICMP flood attack behavior. It also assumes that the extracted traffic features and predefined severity thresholds are adequate to distinguish different attack intensities and map them into meaningful severity levels.

However, the framework is limited to ICMP flood attack analysis and does not currently address all possible DoS or multi-vector attack scenarios. In addition, the assigned ICMP impact weight and severity thresholds may require further refinement when applied in different operational environments. Despite these limitations, the proposed approach provides a structured foundation for combining attack detection with severity-aware prioritization in network security monitoring.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Component Architecture Overview

The proposed ICMP Flood Detection & Severity Analysis component operates as Component 1 within the broader multi-layer DoS detection framework. The system follows an Input–Process–Output pipeline architecture:

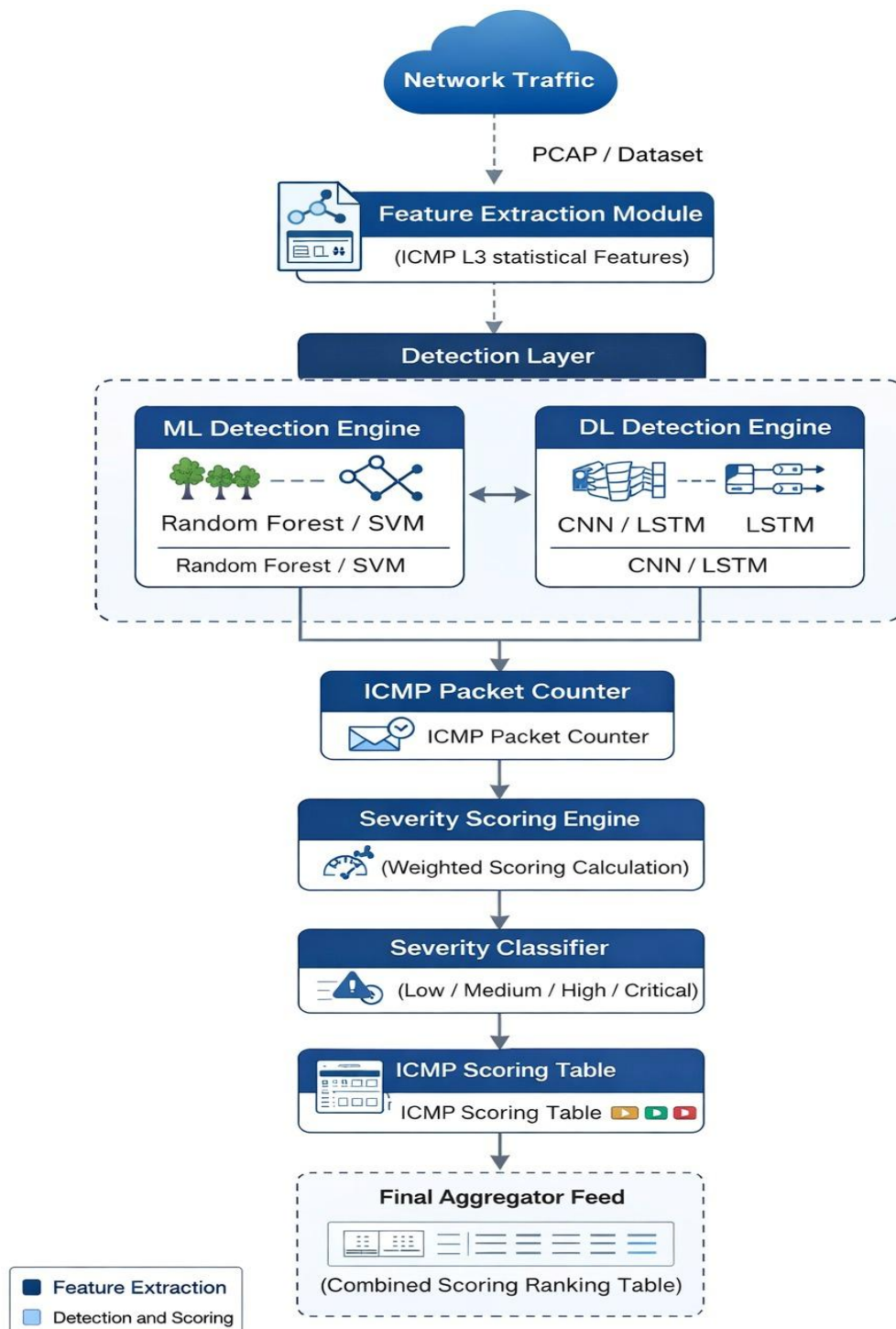


Figure 1 icmp detection system architecture flowchart

Stage	Component	Description
INPUT	Network Traffic Source	CICDDoS2019 / CAIDA datasets; near real-time or offline PCAP traffic
INPUT	Feature Extraction Module	Extracts ICMP-specific L3 features using time-window statistical analysis
PROCESS	ML Detection Engine	Random Forest and SVM classifiers for structured feature classification
PROCESS	DL Detection Engine	CNN/LSTM networks for temporal traffic pattern recognition
PROCESS	ICMP Packet Counter	Counts detected malicious ICMP packets per time window
PROCESS	Severity Scoring Engine	Applies weight 1.5 to packet count; computed severity score
PROCESS	Severity Classifier	Maps score to Low / Medium / High / Critical levels
OUTPUT	ICMP Scoring Table	Structured table with attack type, count, score, severity level
OUTPUT	Final Aggregator Feed	Feeds severity data to the Combined Scoring Ranking Table

Table 4 Input–Process–Output pipeline architecture

4.2 Data Flow

Raw network traffic is ingested and preprocessed to extract ICMP-relevant features. The hybrid ML/DL detection pipeline classifies traffic as attack or normal. Detected attack packets are counted per time window. The severity scoring engine computes a weighted score. The severity classifier maps this to a risk level. The ICMP Scoring Table output is then passed to the Final Severity Aggregator, which combines outputs from all four components (ICMP, SYN, UDP, Fragmentation) into a Combined Scoring Ranking Table for the complete system.

4.3 Scalability and Security Considerations

The system is designed as a prototype-level near real-time framework suitable for experimental evaluation. For scalability, time-window-based analysis allows parallel processing of traffic streams. The design can be extended to production deployment through integration with SDN controllers or SIEM platforms. Security considerations include

ensuring the integrity of feature extraction pipelines and preventing adversarial traffic manipulation that could circumvent detection thresholds.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Stakeholder	Role	Interest
SOC Analysts	Primary Users	Receive severity-ranked alerts to priorities ICMP flood incident response
Network Administrators	Primary Users	Monitor ICMP traffic patterns; configure detection thresholds
IT Security Managers	Decision Makers	Use severity reports to allocate resources and assess risk exposure
Research Supervisor	Academic Oversight	Evaluate technical contribution, novelty, and research quality

Table 5 stakeholders

5.2 Functional Requirements

1. The system shall ingest network traffic data from CICDDoS2019 and CAIDA datasets in PCAP or CSV format for processing.
2. The system shall extract ICMP-specific Layer 3 features including packet rate, echo request count, source IP diversity, and time-window statistics.
3. The system shall classify network traffic as ICMP flood attack or normal traffic using trained ML models (Random Forest, SVM).
4. The system shall classify network traffic using DL models (CNN or LSTM) to capture temporal attack patterns.
5. The system shall count detected malicious ICMP attack packets within defined time windows.
6. The system shall compute a severity score by multiplying the ICMP packet count by the impact weight.
7. The system shall classify the severity score into one of four severity levels: Low, Medium, High, or Critical.
8. The system shall generate a structured ICMP Attack Severity Ranking Table containing attack type, packet count, severity score, and severity level.
9. The system should export severity data to the Final Severity Aggregator for integration with other system components.
10. The system shall log detection results and severity scores for post-analysis and validation.

5.3 Non-Functional Requirements

Category	Requirement
Security	The system shall operate exclusively on anonymized or synthetic datasets; no real attack traffic shall be generated against live systems
Performance	The hybrid ML/DL detection pipeline shall achieve $\geq 90\%$ accuracy and produce severity scores within acceptable processing time for near real-time operation
Usability	The ICMP Severity Ranking Table output shall be human-readable and formatted to support rapid SOC decision-making without requiring specialized data science expertise
Scalability	The time-window-based analysis architecture shall support extension to larger traffic volumes through parallel processing
Reliability	The detection models shall maintain consistent performance across both CICDDoS2019 and CAIDA validation datasets
Maintainability	The severity weight and classification thresholds shall be configurable parameters to allow empirical refinement without requiring code changes

Table 6 Non-Functional Requirements

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target markets for the commercialized system include.

1. Enterprise security operations centers require automated threat prioritization tools.
2. Cloud service providers managing large-scale network infrastructure.
3. Internet Service Providers (ISPs) monitoring volumetric attack traffic.
4. Financial institutions and healthcare networks require SLA-compliant availability guarantees. The customer persona is a mid-to-large enterprise with a dedicated SOC team seeking to reduce mean time to respond (MTTR) to network-layer DoS attacks.

6.2 Value Proposition

The proposed system uniquely combines multi-layer DoS detection with severity quantification a capability absent in current commercial and open-source solutions such as Suricata IDS, Snort, and HAProxy. By enabling risk-prioritized incident response, the system reduces analyst cognitive load, minimizes response time to high-severity ICMP attacks, and supports regulatory compliance in critical infrastructure sectors.

6.3 Revenue Model and Pricing Strategy

- SaaS Subscription Model: Cloud-hosted version with tiered pricing based on monitored traffic volume
- On-Premises Licensing: One-time license fee for deployment within enterprise network security infrastructure
- Research/Academic Licensing: Reduced-cost access for academic institutions and research organizations.

6.4 Competitive Advantage

Existing commercial IDS/IPS solutions provide binary attack detection without severity scoring. The proposed system's weighted severity quantification mechanism provides a differentiated value proposition, particularly for organizations requiring risk-prioritized automated response workflows.

6.5 Intellectual Property Considerations

The severity scoring algorithm, impact weight derivation methodology, and combined ranking architecture represent novel contributions suitable for patent application consideration. Academic publication in IEEE cybersecurity venues is planned to establish prior art and research credibility.

7. BUDGET AND JUSTIFICATION

Total Estimated Budget: LKR 35,000

Category	Item	Estimated Cost (LKR)	Justification
Hardware	High-performance laptop/workstation (if required)	0	Required for training ML/DL models on large datasets (CICDDoS2019 80GB)
Software	Python, TensorFlow, Scikit-learn (open source)	0	All ML/DL tools are freely available open source
Software	Wireshark, Scapy (open source)	0	Open-source network analysis tools
Cloud Services	Google Colab Pro / AWS EC2 (GPU training)	25,000	GPU acceleration required for CNN/LSTM model training on large datasets
Data	CICDDoS2019 / CAIDA dataset access	5,000	Dataset download and storage costs
Human Effort	Research team time (4 members x 6 months)	0	Estimated value of student research effort including analysis and documentation
Miscellaneous	Printing, binding, presentation materials	5000	Final report preparation and submission
	TOTAL	35,000	

Table 7 budget

8.WORK BREAKDOWN STRUCTURE (WBS)

Task ID	Task Description	Assignee	Duration	Deliverable
T1.1	Literature review on ICMP flood detection and ML/DL approaches	KOWSHICK S N	Week 1–3	Annotated bibliography
T1.2	Dataset acquisition and preprocessing (CICDDoS2019, CAIDA)	KOWSHICK S N	Week 2–4	Preprocessed dataset
T1.3	ICMP feature extraction and selection	KOWSHICK S N	Week 4–6	Feature matrix
T1.4	Random Forest and SVM model implementation	KOWSHICK S N	Week 5–8	Trained ML models
T1.5	CNN/LSTM deep learning model implementation	KOWSHICK S N	Week 7–10	Trained DL models
T1.6	ICMP packet counting module development	KOWSHICK S N	Week 9–11	Packet counter module
T1.7	Severity scoring engine implementation	KOWSHICK S N	Week 10–12	Scoring engine
T1.8	Severity classification and ICMP ranking table generation	KOWSHICK S N	Week 12–14	ICMP Scoring Table
T1.9	Model evaluation and validation	KOWSHICK S N	Week 13–16	Evaluation report
T1.10	Integration with Final Severity Aggregator	All team members	Week 15–17	Integrated component
T1.11	Final documentation and report writing	KOWSHICK S N	Week 16–18	Final report

Table 8 work breakdown

9. GANTT CHART

Task	W1-2	W3-4	W5-6	W7-8	W9-10	W11-12	W13-14	W15-16	W17-18
Literature Review	####	####							
Dataset Preprocessing		####	####						
Feature Extraction			####	####					
ML Model (RF/SVM)				####	####				
DL Model (CNN/LSTM)					####	####			
Packet Counter					####	####			
Severity Scoring Engine						####	####		
Severity Classification							####	####	
Integration & Testing								####	####
Documentation							####	####	####

Table 9 Gantt

Key Milestones: Dataset Ready (Week 4) | ML Models Trained (Week 10) | Severity Engine Complete (Week 12) | Component Integrated (Week 16) | Final Report Submitted (Week 18)

10. References

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11. APPENDICES

Appendix A: Risk Analysis

Risk	Likelihood	Impact	Mitigation Strategy
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Dataset does not contain sufficient ICMP flood samples	Low	High	Use both CICDDoS2019 and CAIDA; apply data augmentation if needed
ML model overfitting on training data	Medium	High	Apply k-fold cross-validation; use regularisation techniques
DL model training time exceeds available compute	Medium	Medium	Use Google Colab Pro GPU; reduce model complexity if needed
Severity weight (1.5) does not align with empirical results	Medium	Medium	Iteratively refine weights using experimental attack impact metrics
Integration with other team components delayed	Low	Medium	Define clear API interface early; use modular design

Table 10 risk analysis

Appendix B: ICMP Packet Feature List

Feature Name	Description	Type
icmp_pkt_rate	Number of ICMP packets per second in time window	Numerical
icmp_echo_req_count	Total ICMP Echo Request packets in window	Numerical
icmp_echo_rep_count	Total ICMP Echo Reply packets in window	Numerical
src_ip_unique_count	Number of unique source IP addresses	Numerical
avg_pkt_size	Average ICMP packet size in bytes	Numerical
inter_arrival_time_mean	Mean inter-arrival time between ICMP packets	Numerical
inter_arrival_time_std	Standard deviation of inter-arrival time	Numerical
icmp_type_code	ICMP type and code field values	Categorical
ttl_mean	Mean TTL value of ICMP packets	Numerical
label	Ground truth: 0 = Normal, 1 = ICMP Flood Attack	Binary

Table 11 icmp packet feature list

Appendix C: Severity Score Calculation Example

Parameter	Value
Detected ICMP Attack Packets	3,500
Impact Weight	1.5
Computed Severity Score	$3,500 \times 1.5 = 5,250$
Severity Level	Critical (Score > 3,000)
Recommended Action	Immediate response required — block source IP, alert SOC manager

Table 12 severity score